

AI-Assisted Job Discovery & Resume Tailoring System

PRD · System Design · Tech Stack

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1 1. Executive Summary

A personal tool that, every morning, finds new relevant job postings across India, tailors your resume to each one, and hands you a direct apply link — you review and click apply yourself, nothing is auto-submitted. Designed to run entirely on free-tier infrastructure so it never breaks or degrades because a paid subscription lapsed, and never surfaces stale postings.

2 2. Problem Statement

Manually searching multiple job boards daily, judging fit against your own resume, and rewriting that resume per application is slow, repetitive, and error-prone under placement-season time pressure. Fully-automated “auto-apply” tools solve the speed problem but create two new ones: they violate most platforms’ Terms of Service (risking account bans at the worst possible time), and content changes to a Word-based resume routinely break formatting/alignment.

3 3. Goals and Non-Goals

3.1 3.1 Goals

- Daily surfacing of new, relevant India job postings matched against your actual resume content.
- Per-job tailored resume: reworded/reordered **existing** content only — never invented skills, numbers, or experience.
- Consistent, professional PDF output regardless of how much text changes, via LaTeX templating.
- Every application ends in a manual click on a direct link — zero automated submission, zero ToS risk.
- Zero recurring monetary cost; nothing in the pipeline requires an active paid subscription to function correctly.

3.2 3.2 Non-Goals

- No automated form-filling or submission on third-party platforms.
- Not an interview-prep or networking tool.
- Not attempting full job-board coverage on day one — start narrow (1–2 sources), expand deliberately.
- Not a resume *writer* from scratch — it tailors an existing, already-good resume.

4 4. User & Asset Context

Asset you have	Role in this system
ChatGPT Go subscription	Interactive chat only — no API credits included , not usable by the automated pipeline
Gemini Pro (Google One AI)	App-only — no API credits included , but a separate free Google AI Studio API key can be created independent of this subscription
No Claude Pro / API	Not required — every component below is chosen to avoid needing this

Asset you have	Role in this system
Existing resume (clean, LaTeX-friendly layout)	Becomes the structured content source <i>and</i> the visual template baseline
Local machine with <code>pdflatex</code> / <code>xelatex</code>	Confirmed available — PDFs compile locally, no paid LaTeX service
Target roles/companies	SDE roles; Amazon and Goldman Sachs named as off-campus targets, alongside on-campus drives

Key implication: the money you already pay for ChatGPT Go / Gemini Pro doesn’t transfer to API automation use. You’ll separately create a free Google AI Studio API key — this costs nothing extra and is a five-minute signup, distinct from the Gemini Pro app subscription.

5 5. Core User Flow

1. You (or a schedule) trigger the daily run.
2. System pulls new postings matching your saved search profile (title / keywords / location).
3. Postings are deduplicated against previously-seen jobs and filtered by a freshness cutoff.
4. Remaining postings are ranked locally against your resume for relevance.
5. Top matches go through per-job resume tailoring.
6. Tailored content is rendered into your LaTeX template and compiled to PDF.
7. You receive a digest: job title, company, direct apply link, tailored PDF, and a short diff of what changed.
8. **You review and click apply yourself.**

6 6. Functional Requirements

ID	Requirement
FR1	Search jobs by keyword + location (India-wide or city-specific)
FR2	Persist “already seen” job IDs to avoid repeat alerts
FR3	Drop postings older than a configurable freshness window (default 21 days)
FR4	Extract resume content once into structured JSON — the single source of truth
FR5	Per-job tailoring reorders/rewords FR4’s JSON only; no new claims introduced
FR6	Render JSON → LaTeX → PDF matching the existing visual template exactly
FR7	Deliver a daily digest with direct apply links and tailored PDFs
FR8	Degrade gracefully (not silently wrong) if the LLM API quota is exhausted
FR9	Provide a short, human-readable diff of what the tailoring step changed
FR10	Store resume content and job history locally — no third-party data sharing

7 7. Non-Functional Requirements

Category	Requirement
Cost	Rs. 0 recurring cost under normal personal-use volume (tens of jobs/day)
Reliability	A missing/exhausted API key degrades functionality, never crashes the run
Freshness	No posting older than the configured window is ever surfaced
Formatting	Tailored PDF must be visually consistent with the original across all content variations
Privacy	Resume and job history stay on your machine; API calls send only the minimum needed text
Auditability	Every tailored resume's changes must be diffable against the original for a quick sanity check before applying
Portability	Runs on a personal laptop; no dependency on a specific paid cloud account

8. System Architecture

8.1 Job Source Layer

Primary source: **Adzuna API**, which has India coverage and a free tier (roughly 1,000 calls/month, about 33/day at time of writing) — comfortably enough for a daily personal digest. Returns title, company, location, posted date, and a direct redirect URL.

Secondary (later phase): individual company career-page feeds for large employers (e.g., Amazon, Goldman Sachs) that don't always route through aggregators.

Job-board APIs change quotas and terms over time — verify current limits at build time rather than relying solely on this document.

8.2 Dedup & Freshness Filter

A local store (flat JSON file or SQLite) of job IDs already seen. Every posting's `posted_date` is checked against a configurable cutoff (default 21 days) before it's allowed further into the pipeline — freshness is enforced as a hard filter, not a sort order.

8.3 Relevance Ranking

Runs **locally**, no API cost: `sentence-transformers` embeds the job description and your resume JSON, scores similarity, and only postings above a threshold proceed to tailoring. This is what keeps LLM API usage low enough to comfortably fit inside any free tier.

8.4 LLM Tailoring Layer

Google AI Studio API (free tier), Flash-class model, called with structured JSON in and JSON out (not free text) — this makes it straightforward to validate that no field contains content absent from the source resume JSON.

8.5 Quota Check & Fallback

Before each call, the pipeline checks whether the day's free quota is likely exhausted. If so, it **does not fail** — it falls back to a local keyword-overlap reordering of your existing bullets (no rewriting, no LLM call). You still get a usable, correctly-formatted tailored PDF; you just don't get LLM-level rewording that day. This is the mechanism that prevents “breaks without a subscription.”

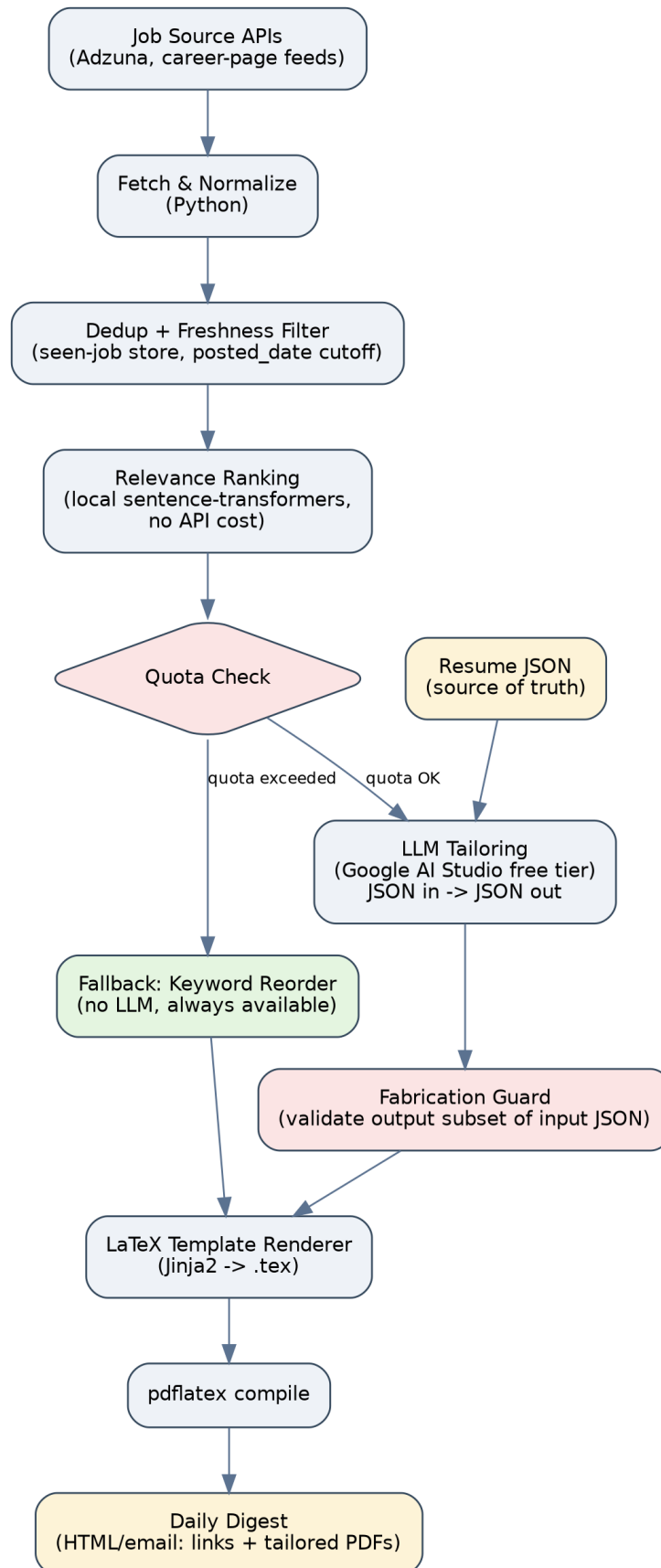


Figure 1: System architecture

8.6 8.6 Fabrication Guard

After tailoring, every output field is checked against the source resume JSON (e.g., new proper nouns, numbers, or skill terms not present in the source trigger a rejection and fallback to the keyword-reorder path instead of silently shipping an inaccurate resume).

8.7 8.7 Resume Rendering

A LaTeX template built from your current resume's layout, with Jinja2 placeholders per JSON field. Same template + new JSON → pdf_lat_ex → PDF, every time — this is what guarantees identical formatting regardless of content length changes.

8.8 8.8 Delivery

An HTML digest (and/or email via Gmail SMTP with an app password — free, no paid service) listing each job with its tailored PDF and direct apply link.

9 9. Data Schemas

9.1 9.1 Resume JSON (source of truth)

```
{
  "profile_summary": "string",
  "education": [
    {"institution": "string", "degree": "string", "duration": "string", "details": "string"}
  ],
  "projects": [
    {
      "name": "string",
      "stack": "string",
      "bullets": ["string", "string"],
      "links": {"demo": "url", "repo": "url"}
    }
  ],
  "skills": {"languages": ["string"], "frameworks": ["string"], "tools": ["string"]},
  "certifications": ["string"]
}
```

9.2 9.2 Job Posting (normalized)

```
{
  "job_id": "string",
  "title": "string",
  "company": "string",
  "location": "string",
  "posted_date": "ISO-8601 date",
  "description": "string",
  "apply_url": "string",
  "source": "adzuna | career_page"
}
```

9.3 9.3 Tailoring Output (validated against 9.1 before use)

Same shape as 9.1, values only reordered/reworded — any key present in this output but absent (as a concept) from the source JSON fails the fabrication guard (§8.6).

10 10. Tech Stack Summary

Layer	Choice	Cost
Job data	Adzuna API	Free tier
Ranking/matching	sentence-transformers (local)	Free
Tailoring LLM	Google AI Studio API (separate key from Gemini Pro app)	Free tier
Resume templating	LaTeX (pdf _l at _e x, Jinja2 for placeholders)	Free
Local storage	JSON file or SQLite (seen-jobs, resume source)	Free
Scheduling	Cron (local) or GitHub Actions (free minutes)	Free
Delivery	Gmail SMTP app password	Free
Language/libraries	Python — requests, jinja2, sentence-transformers	Free

Net recurring cost: Rs. 0. ChatGPT Go and Gemini Pro app subscriptions remain purely for your own interactive use and are not load-bearing for this system.

11 11. Security & Privacy

- Resume content and job history are stored locally, never uploaded to a third party beyond the minimum text sent to the job-search and LLM APIs for a single request.
- API keys (Adzuna, Google AI Studio) are kept in a local `.env` file, excluded from any version control (`.gitignore`), never hardcoded.
- Google AI Studio’s free tier may use submitted data to improve Google’s models — avoid sending anything beyond resume/job-description text (no personal ID numbers, no unrelated personal data).
- Gmail app passwords are scoped to SMTP sending only, not full account access.

12 12. Cost Analysis

Item	Free-tier ceiling	Expected personal usage	Headroom
Adzuna calls	~33/day	1–3/day (one search profile, run once daily)	Large
Google AI Studio calls	Low tens to low hundreds/day (model-dependent, changes over time)	5–15/day (only top-ranked matches get tailored)	Comfortable
Local embedding ranking	N/A (offline)	Unlimited	N/A
LaTeX compilation	N/A (offline)	Unlimited	N/A

Conclusion: normal personal use sits well inside every free tier; the fallback path (§8.5) exists for the rare day it doesn’t.

13 13. Build Phases

Phase 1 — Resume foundation Extract current resume into the JSON schema (§9.1); build the LaTeX template; confirm the compiled PDF is visually identical to the original.

Phase 2 — Job ingestion Adzuna integration; dedup store; freshness filter; manual run printing results to console (no tailoring yet).

Phase 3 — Relevance ranking Local sentence-transformers scoring of postings against resume JSON; threshold tuning.

Phase 4 — Tailoring + guardrails Google AI Studio integration; JSON-in/JSON-out prompting; fabrication guard (§8.6); keyword-only fallback path (§8.5).

Phase 5 — Delivery & scheduling Digest generation (HTML/email); cron or GitHub Actions scheduling; end-to-end dry run.

14 14. Testing & Validation Plan

Test	Purpose
Fabrication guard unit tests	Feed tailoring output containing an invented skill; confirm it's rejected and falls back correctly
Freshness filter test	Feed a posting older than the cutoff; confirm it never reaches the digest
Dedup test	Run the pipeline twice on the same data; confirm no duplicate alert
Quota-exhaustion simulation	Force the fallback path; confirm a usable PDF still results
Visual regression	Compare rendered PDF against the original resume for layout drift after multiple tailoring cycles

15 15. Risk Register

Risk	Likelihood	Mitigation
Job-board API terms/quota change	Medium	Verify limits at build time; keep source layer swappable
LLM free-tier quota cut further	Medium	Fallback path (§8.5) already assumes this will happen
Tailoring subtly overstates experience	Low–Medium	Fabrication guard (§8.6) plus your own manual review before applying
Large employers (Amazon, Goldman Sachs) under-represented by aggregators	Medium	Phase 5+: add direct career-page feeds for named targets
Local script stops running (laptop off, forgot to trigger)	Medium	Optional GitHub Actions scheduling removes the “must remember” dependency

16 16. Success Metrics

- Daily digest delivered with zero duplicate or stale postings.
- Tailored PDF formatting indistinguishable from the original resume, regardless of content length.

- Rs. 0 recurring cost sustained over a full month of normal use.
- Time from “new relevant posting” to “tailored resume + link in hand” under a few minutes.

17 17. Setup Checklist (Appendix)

1. Register a free Adzuna developer account → obtain `app_id` and `app_key`.
2. Create a Google AI Studio API key (separate from the Gemini Pro app subscription) → store as an environment variable.
3. Confirm `pdflatex/xelatex` installed locally (already verified working).
4. Set up a Gmail app password for digest delivery (optional if using a local HTML file instead).
5. Initialize local storage (JSON/SQLite) for seen-job IDs and the resume JSON source of truth.
6. (Optional, later) Set up a GitHub Actions workflow for unattended daily scheduling.

18 18. Open Questions

- Which company career pages beyond Adzuna’s coverage matter enough to add individually?
- Preferred delivery format: email digest vs. a local webpage opened each morning?
- Default search profile: one broad SDE search, or several role-specific ones (SDE / ML / backend)?